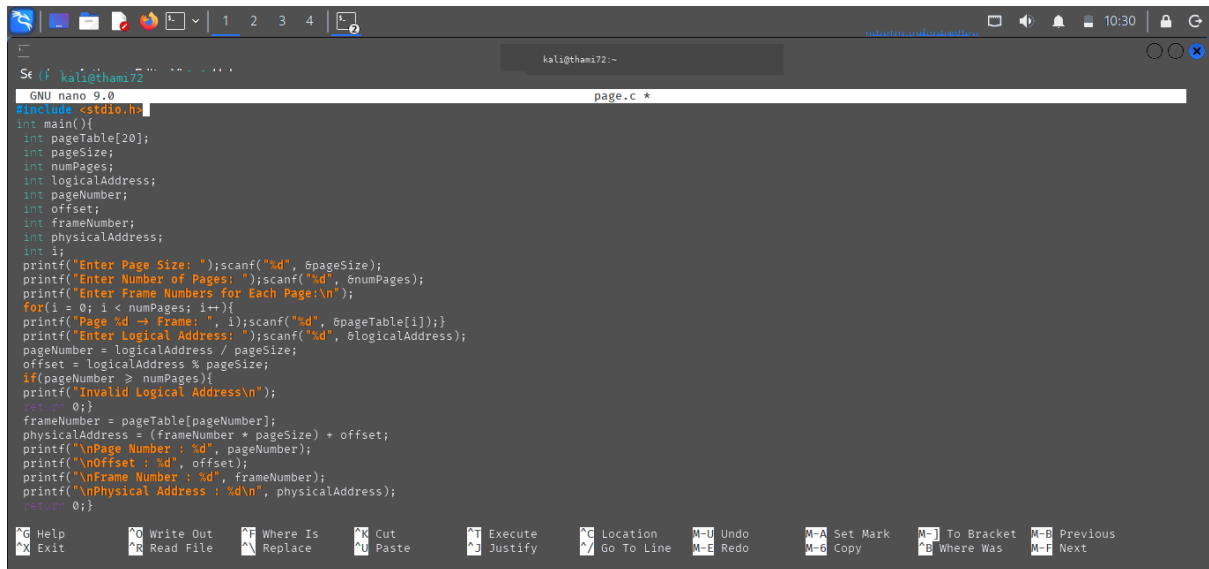
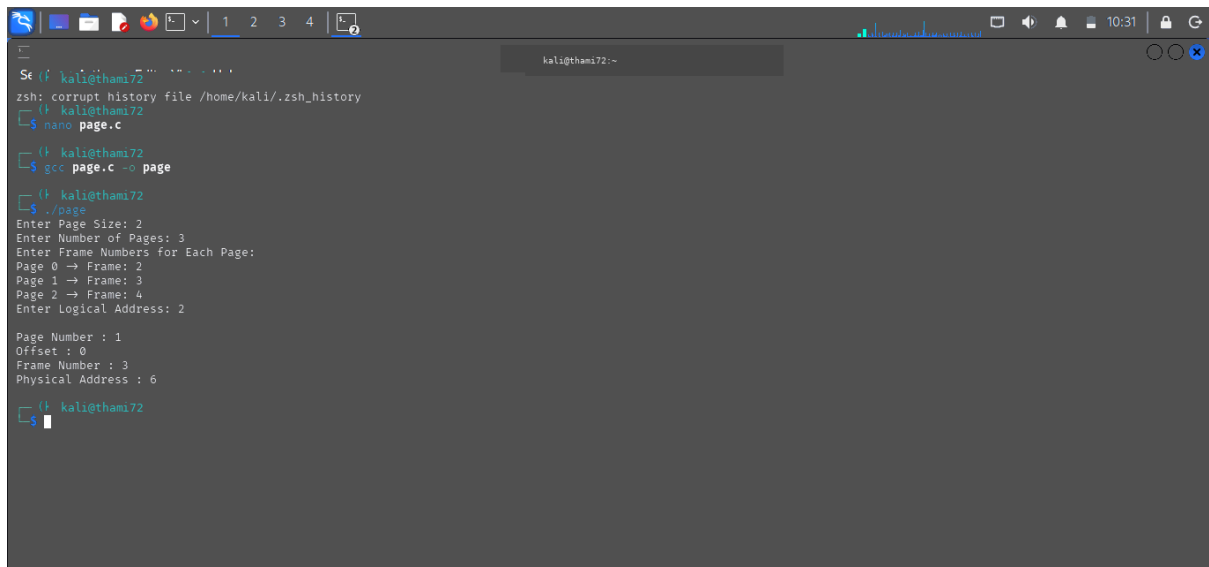


C PROGRAMMING :PAGING



```
GNU nano 9.0 page.c *
#include <stdio.h>
int main(){
    int pageTable[20];
    int pageSize;
    int numPages;
    int logicalAddress;
    int pageNumber;
    int offset;
    int frameNumber;
    int physicalAddress;
    int i;
    printf("Enter Page Size: ");scanf("%d", &pageSize);
    printf("Enter Number of Pages: ");scanf("%d", &numPages);
    printf("Enter Frame Numbers for Each Page:\n");
    for(i = 0; i < numPages; i++){
        printf("Page %d -> Frame: ", i);scanf("%d", &pageTable[i]);
    }
    printf("Enter Logical Address: ");scanf("%d", &logicalAddress);
    pageNumber = logicalAddress / pageSize;
    offset = logicalAddress % pageSize;
    if(pageNumber >= numPages){
        printf("Invalid Logical Address\n");
        return 0;
    }
    frameNumber = pageTable[pageNumber];
    physicalAddress = (frameNumber * pageSize) + offset;
    printf("\nPage Number : %d", pageNumber);
    printf("\nOffset : %d", offset);
    printf("\nFrame Number : %d", frameNumber);
    printf("\nPhysical Address : %d\n", physicalAddress);
    return 0;
}
```

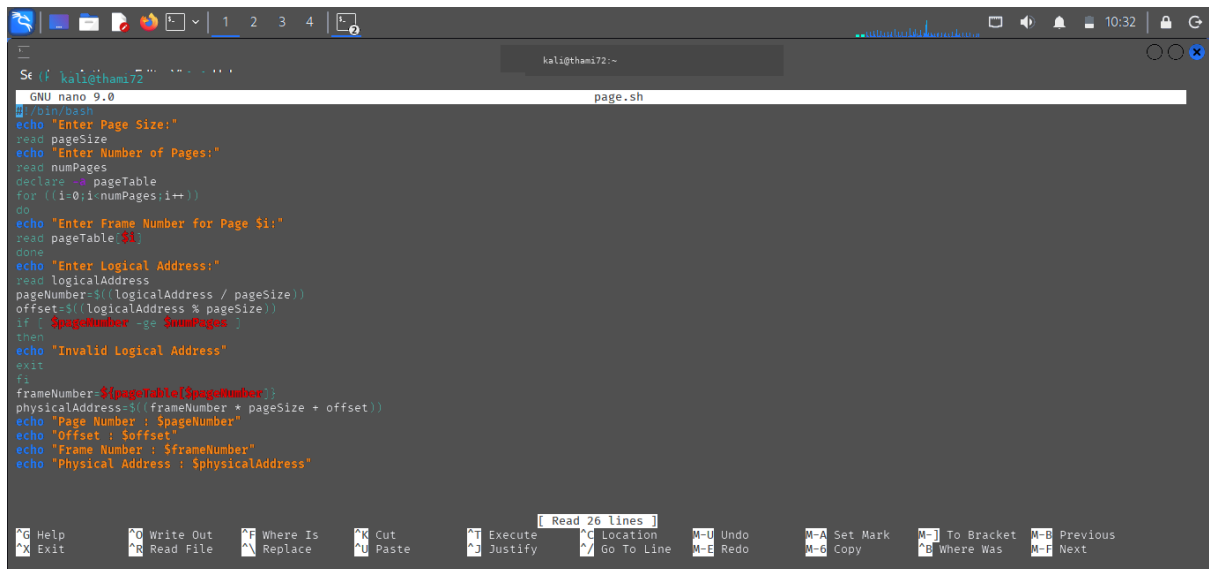
OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano page.c
kali@thami72 ~
$ gcc page.c -o page
kali@thami72 ~
$ ./page
Enter Page Size: 2
Enter Number of Pages: 3
Enter Frame Numbers for Each Page:
Page 0 -> Frame: 2
Page 1 -> Frame: 3
Page 2 -> Frame: 4
Enter Logical Address: 2

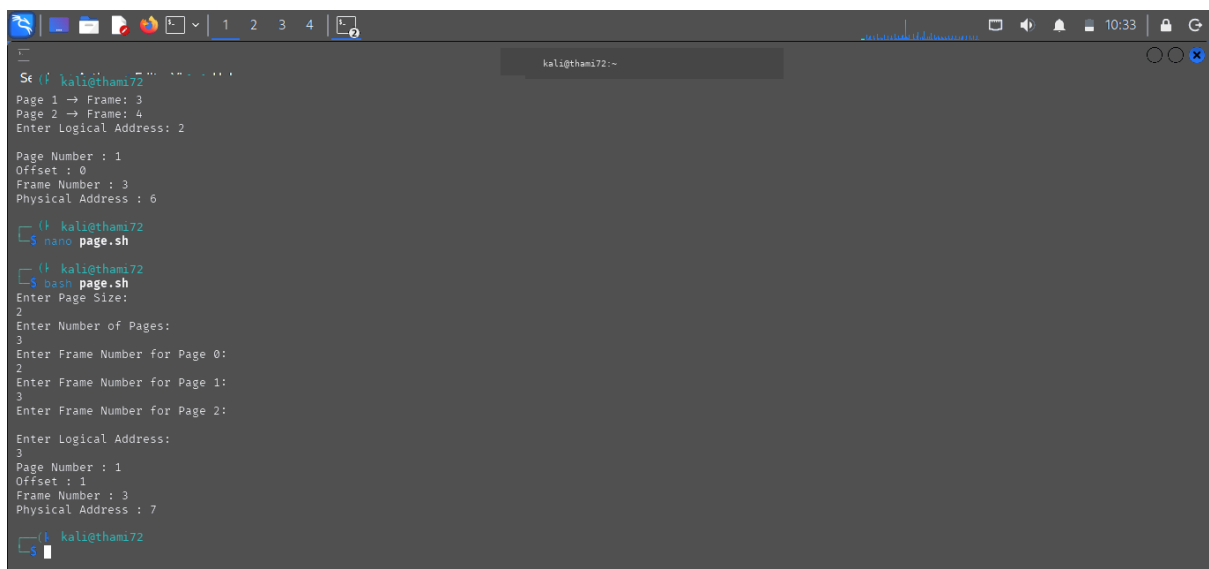
Page Number : 1
Offset : 0
Frame Number : 3
Physical Address : 6
kali@thami72 ~
$
```

SHELL PROGRAMMING : PAGING



```
GNU nano 9.0 page.sh
#!/bin/bash
echo "Enter Page Size:"
read pageSize
echo "Enter Number of Pages:"
read numPages
declare -a pageTable
for ((i=0;i<numPages;i++))
do
echo "Enter Frame Number for Page $i:"
read pageTable[i]
done
echo "Enter Logical Address:"
read logicalAddress
pageNumber=$((logicalAddress / pageSize))
offset=$((logicalAddress % pageSize))
if [ $pageNumber -ge $numPages ]
then
echo "Invalid Logical Address"
exit
fi
frameNumber=${pageTable[$pageNumber]}
physicalAddress=$((frameNumber * pageSize + offset))
echo "Page Number : $pageNumber"
echo "Offset : $offset"
echo "Frame Number : $frameNumber"
echo "Physical Address : $physicalAddress"
```

OUTPUT :



```
Se (f kali@thami72 ~)
Page 1 -> Frame: 3
Page 2 -> Frame: 4
Enter Logical Address: 2

Page Number : 1
Offset : 0
Frame Number : 3
Physical Address : 6

kali@thami72 ~$ nano page.sh
kali@thami72 ~$ bash page.sh
Enter Page Size:
2
Enter Number of Pages:
3
Enter Frame Number for Page 0:
2
Enter Frame Number for Page 1:
3
Enter Frame Number for Page 2:
3
Enter Logical Address:
3
Page Number : 1
Offset : 1
Frame Number : 3
Physical Address : 7

kali@thami72 ~$
```